

UNIGE Mobile Robots exam - September 9, 2022 - 2 h
 Work on paper, not on a tablet. Use a pen with good contrast (no pencil).

Exercise 1. Parameterization (10 min)

Parameterize the following robot and **give the configuration vector**. Wheels 3 and 4 are rigidly linked. Still, they can be represented as two general wheels. In the parameter table, the fact that the wheels are rigidly linked will be reflected by **replacing β_4 by its expression as a function of β_3** (compulsory: indicating just β_4 will be graded as wrong.).

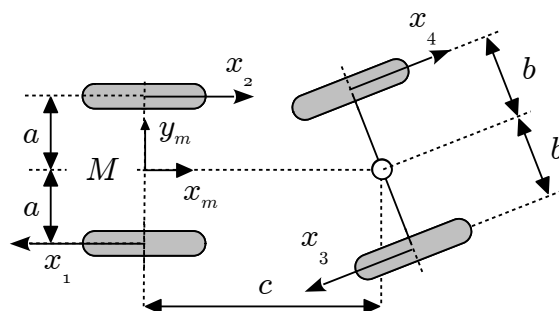
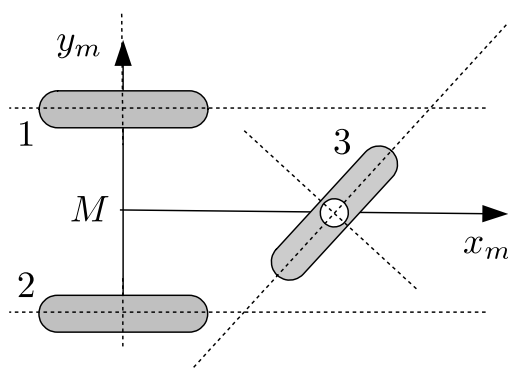


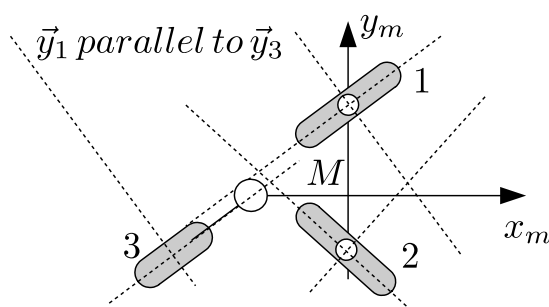
Figure 1.1. Robot to parameterize

Exercise 2. Kinematics related short questions (30 min)

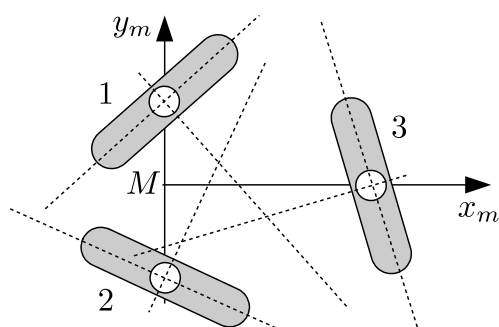
2.1 to 2.4: For each vehicle, give the position of the Instantaneous Center of Rotation if it can be determined or state that it cannot be determined. In each case, a short justification is expected. In your solution/explanation, use the standard point names ($A_1, B_1, \dots, M$) and vector names ($\vec{x}_1, \vec{y}_1, \dots$). You are allowed to use $\vec{y}_1$ as a shorthand for “the line through B_1 directed by $\vec{y}_1$ ”, and define a location as “the intersection of $\vec{y}_1$ and $\vec{y}_2$ ”. Properly use such methods and **do not use sketches (drawings)**.



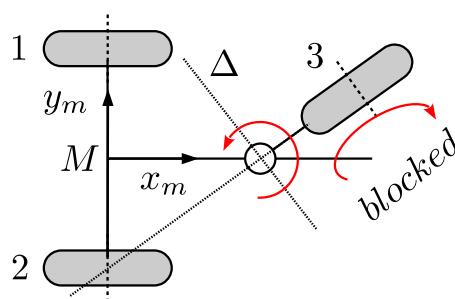
2.1.



2.2.



2.3.



2.4

2.5. to 2.6: For both questions, the problem is to justify the standard wheel configuration (**not** the motor configuration) from the robot type, without calculations and without using the following inequations $0 \leq \delta_s + \delta_f \leq 2$ and $2 \leq \delta_m + \delta_s \leq 3$. You will assume that the robot has three wheels.

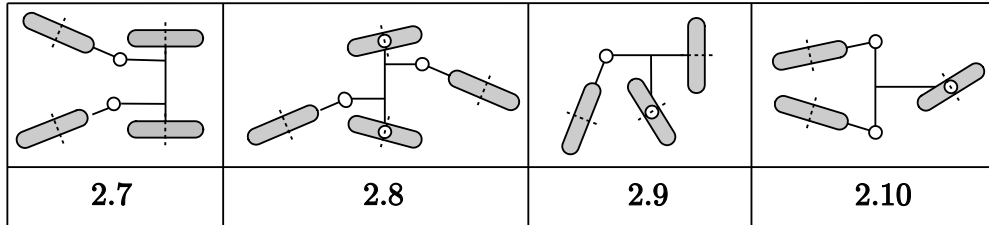
Beware! The problem is to **deduce** the three wheels **from** (δ_m, δ_s) , and **not** the other way around. Given the type, you need to prove that the wheels **have to be** those of the standard configuration, either because there is no other solution, or because other solutions are inferior. If you reason the other way around (take a wheel configuration and show that it does lead to the correct type), your answer will be graded as zero.

In both cases, clarity and conciseness will be evaluation criteria, and the answer must not exceed ten lines of text.

2.5. Justify the standard wheel configuration of the (1,1) robot.

2.6. Justify the standard wheel configuration of the (2,0) robot.

2.7 to 2.10: Give the type of each of the robots, with a short and clear justification. Clarity and conciseness will be important evaluation criteria, typically 2/3 of the grade.



Exercise 3. Kinematic modelling (50 min)

We are going to study the robot of figure 3.1. The study is for a client, who imposes the definition of the robot frame, and of the wheel orientation angles. At the beginning of the exercise, you will be required to find the relation between the client's angles and the standard angles β . **It is not a difficult question, but failure to solve it essentially ruins the whole exercise.** If you really do not know how to solve that question, you may go on with the standard angles used in the class, but there will be a **significant penalty**.

In practice, the robot would typically have at least one more wheel. A simple possibility is a castor wheel without any actuator, that is why **it is completely ignored in this exercise**.

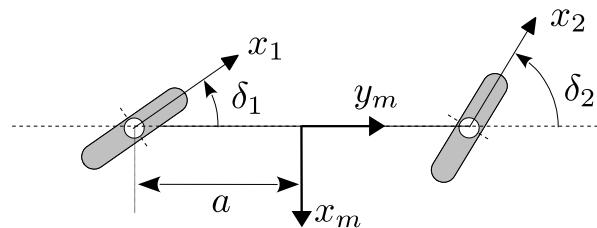


Figure 3.1. The robot, with relevant notations. Beware of the definition of R_m .

- **3.1.** Give the parameter table of the robot. In the table, replace β_1 and β_2 by their expression as functions of δ_1 and δ_2 respectively. Also give the configuration vector.
- **3.2.** Calculate matrices J_1 , J_2 , C_1 and C_2 and C_1^* .
- **3.3.** By considering the determinants of the three minors of C_1^* , prove that its rank is two, except for a certain orientation of the two steerable wheels. Give this orientation.

Recall: the minors of a 2×3 matrix are the three 2×2 matrices obtained by suppressing one column. The matrix has rank less than 2, if and only if the determinants of the three minors are all equal to zero.

Beware you have to **rigorously** prove that the rank is not 2 **if and only if** the values of the angles are a certain set of values. I insist on the “only if” part. A mere statement like “The rank is 2 except for $\delta_1 = 0$ and $\delta_2 = \pi/2$ ” without a rigorous proof will not be accepted.

- **3.4.** Except in this particular wheel orientation, what is the type of the robot?
- **3.5** Assuming the orientation of the steerable wheels are blocked at the above mentioned orientation (they become fixed wheels), what does the robot type become?
- **3.6.** Assuming the robot is never in this particular configuration, give a base of $\text{Ker}(C_1^*)$. Again, do not just give the base: clearly prove that it is a base.
- **3.7.** Give the posture kinematic model of the robot.

Tip: To solve this kind of system:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0 \end{cases}$$

Replace equation 2 by a linear combination of equations 1 and 2 to put the system under an equivalent triangular form, which makes it easy to find a vector that satisfies both equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0 \\ a'_{22}x_2 + a'_{23}x_3 = 0 \end{cases}$$

Exercise 4. Kalman filter and observability (30 min)

We study a localization system based on the detection of the transitions between the squares of a checkerboard floor using infrared sensors. In essence, the output of the sensors over a light/dark surface is 1/0. So changes of sensor state correspond to dark-to-bright or bright-to-dark transitions. The robot uses two sensors S_1 and S_2 located at know positions in R_m (see figure 4.1.a).

The transition lines of the checkerboard floor are the beacons of the localization system. The floor tessellation is assumed square, as seen on figure 4.1.b, so the transition is either a line $x = n \times \Delta$ or a line $y = n \times \Delta$. As usual, when establishing the measurement equations we assume that both n and Δ are known. In other words, we assume that we know exactly which transition has been crossed.

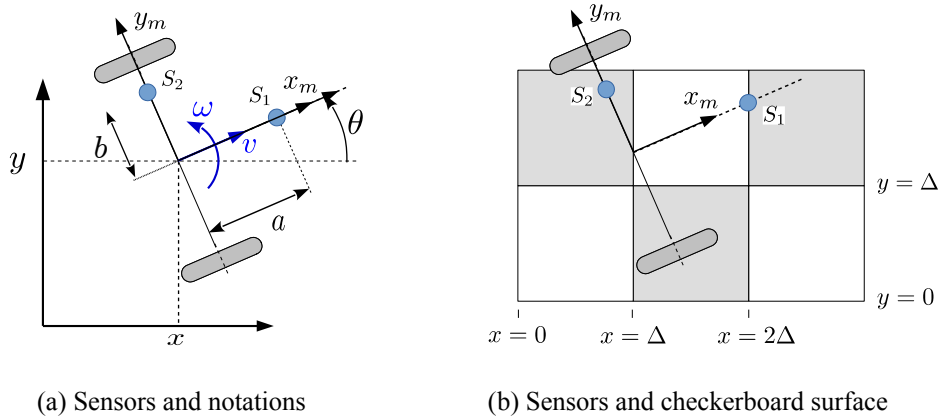


Figure 4.1

4.1. What are the coordinates of the sensors in R_m , mS_1 and mS_2 ?

4.2. When the robot is at $(x, y, \theta)^t$, what are the coordinates of the sensors in R_o , oS_1 and oS_2 ?

4.3. When S_1 crosses a “vertical” line $x=n\Delta$, what is the value of the x coordinate of S_1 in R_o ?

4.4. The state vector is $X = (x, y, \theta)^T$. Δ is a known constant.

a. Write the following observation equations:

- g_{1x} : Corresponding to sensor S_1 crossing a “vertical” line ($x=n\Delta$, n known constant).
- g_{1y} : Corresponding to sensor S_1 crossing a “horizontal” line ($y=n\Delta$, n known constant).
- g_{2x} : Corresponding to sensor S_2 crossing a “vertical” line ($x=n\Delta$, n known constant).
- g_{2y} : Corresponding to sensor S_2 crossing a “horizontal” line ($y=n\Delta$, n known constant).

b. Write the C matrices of the Kalman filter corresponding to these measurement equations, which we will denote as C_{1x} , C_{1y} , C_{2x} , C_{2y} .

4.5. Give the continuous posture kinematic model of the robot. It is a classical one, you do not need to recalculate it or justify it.

4.6. It is easy to realize that, when the robot has its two sensors, the state is observable. Assume now that the robot has only sensor S_2 (e.g. S_1 is defective).

4.6.a. Assuming $b = 0$, find without calculation, a **non-zero** input such that the state is not observable. Justify rapidly.

4.6.b. Is there such a situation when $b \neq 0$? If yes, find it without calculation (defining the corresponding robot motion is enough).

4.6.c. Establish the previous result by calculation.